CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

1.1 Physical Quantities

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Multiple Choice (Paper 1) **Total:** 17 Questions (17 Marks)

Question 1 Easy [1 mark]

Which definition is correct and uses only quantities rather than units?

A Density is mass per cubic metre

B Potential difference is energy per unit current

C Pressure is force per unit area

D Speed is distance travelled per second

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The question asks you to select the correct statement which uses quantities only
- Force and area are both quantities, not units

A is incorrect as while the statement is correct, a metre is a unit

B is incorrect as the statement is incorrect, potential difference is the energy per unit charge

D is incorrect as while the statement is correct, second is a unit

Question 2 Easy [1 mark]

What is a reasonable estimate for the gravitational force acting on an adult person standing on the Earth?

A 50 N

B 300 N

C 400 N

D 700 N

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Gravitational force, or weight, is given by $W = mg$
- On Earth, $g \approx 10 \text{ N kg}^{-1}$
- The mass of a person on Earth is
  - $m = \frac{W}{g}$
  - $m \approx \frac{700}{10} \approx 70 \text{ kg}$
- 70 kg is a reasonable estimate for the mass of an adult person

A is incorrect as $m \approx \frac{50}{10} \approx 5 \text{ kg}$ is not a reasonable estimate for the mass of an adult person

B is incorrect as $m \approx \frac{300}{10} \approx 30 \text{ kg}$ is not a reasonable estimate for the mass of an adult person

C is incorrect as $m \approx \frac{400}{10} \approx 40 \text{ kg}$ is not a reasonable estimate for the mass of an adult person

Question 3 Easy [1 mark]

A physical quantity has the symbol V and the unit m^3.

Which physical quantity does V represent?

A Potential difference

B Velocity

C Pressure

D Volume

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Volume has the symbol V and its units are m^3

A is incorrect because the symbol for potential difference is V, but its units are V

B is incorrect because the symbol for velocity is v and its units are $m\ s^{-1}$

C is incorrect because the symbol for pressure is p and its units are $N\ m^{-2}$ or Pa

Question 4 Easy [1 mark]

What is a reasonable estimate of the number of seconds in a year?

A 5×10^5 s

B 1×10^6 s

C 3×10^7 s

D 7×10^8 s

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- A reasonable estimate for the number of seconds in a year is 365 days × 24 hours × 60 min × 60 s
- $365 \times 24 \times 60 \times 60 = 31\,536\,000 = 3 \times 10^7$ s (1 s.f.)

This is a really important estimate that you need to use a lot in calculations. Always use (365 × 24 × 60 × 60) as the estimate, not the rounded figure.

Question 5 Easy [1 mark]

A physical quantity has the unit m s^{-2} and the symbol a.

What is the correct name for this quantity?

A Activity of a radioactive source

B Amplitude

C Avogadro constant

D Acceleration

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Acceleration has the symbol a and the unit m s^{-2}

A is incorrect because the activity of a radioactive source has the symbol A and the unit Bq

B is incorrect because amplitude has the symbol x_0 and the unit m

C is incorrect because the Avogadro constant has the symbol N_A and the unit mol^{-1}

Question 6 Easy [1 mark]

Which of the following must all physical quantities have?

A Direction and magnitude

B Direction and a unit

C Magnitude and a prefix

D Magnitude and a unit

Mark Scheme & Explanation:

The correct answer is D

- All physical quantities must have a magnitude and a unit

Remember, it is vector quantities that must have both a magnitude and direction.

Question 7 Medium [1 mark]

A student creates a table to show reasonable estimates of some physical quantities.

Which row is a reasonable estimate?

	quantity	value
A	frequency of a radio wave	500 pHz
B	mass of an atom	500 μg
C	the Young modulus of a metal	500 kPa
D	wavelength of green light	500 nm

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The range of visible light is about 400 – 700 nm, so 500 nm is a reasonable estimate for the wavelength of green light
- You should have a rough idea of the ranges of each section of the electromagnetic spectrum

A is incorrect as with a frequency of 500 pHz (500×10^{-12}), this would equate to a wavelength of $\lambda = \frac{v}{f} = \frac{3 \times 10^{8}}{500 \times 10^{-12}} = 6 \times 10^{17}$ m. Typical radio wavelengths range from 1 mm to 100 km, so the estimated value is far too high.

B is incorrect as in kg, 500μg works out as $500 \times 10^{-6} \times 10^{-3} = 5 \times 10^{-7}$ kg. The mass of a proton is 1.67×10^{-27} kg, so the estimated value is far too high

C is incorrect as typical values for the Young Modulus of a metal usually range from 50 to 400 GPa, so the estimated value is far too low.

Question 8 Medium [1 mark]

What is a reasonable estimate for the volume of 1 ml of water?

A $0.1\ cm^3$

B $1\ cm^3$

C $10\ cm^3$

D $100\ cm^3$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- 1 ml = 1 cm^3
- This estimated equivalence is used when determining the density of irregularly shaped objects
- The displaced water in ml is equal to the volume of the object in cm^3

Question 9 Medium [1 mark]

Which row shows what all physical quantities must have?

	magnitude	direction	unit
A	✗	✗	✓
B	✓	✗	✓
C	✓	✓	✗
D	✓	✓	✓

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- All physical quantities must have
  - a magnitude
  - a unit
- They don't have to have a direction, only vector quantities do

Question 10 Medium [1 mark]

An adult male stands with a litre carton of milk balanced on his head.

What is the best estimate of the gravitational potential energy of the milk carton?

A 2 J

B 20 J

C 200 J

D 2000 J

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- A reasonable estimate for the mass of a litre carton on milk is 1 kg
- A reasonable estimate of the gravitational field strength on Earth is 10 N kg^{-1}
- A reasonable estimate for the height of an adult male is 2 m
- Therefore, a reasonable estimate for the gravitational potential energy of the milk carton is:
  - $E_p = mg\Delta h$
  - $E_p \approx 1 \times 2 \times 10 \approx 20$ J

Question 11 Medium [1 mark]

Which of the following is a reasonable estimate for the density of a sugar cube?

A 0.01 kg m^{-3}

B 0.1 kg m^{-3}

C 10 kg m^{-3}

D 1000 kg m^{-3}

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- A reasonable estimate for the mass of a sugar cube is 1 g = 1 × 10^{-3} kg
- A reasonable estimate for the volume of a sugar cube is 1 cm^3 = 1 × 10^{-6} m^3
- Therefore a reasonable estimate for the density of a sugar cube is:
  - $\rho = \frac{m}{V}$
  - $\rho \approx \frac{1 \times 10^{-3}}{1 \times 10^{-6}} \approx 1000 \text{ kg m}^{-3}$

Converting cm^3 to m^3 is a conversion that you need to be familiar with.

- 1 m = 100 cm
- Therefore, 1 m^3 = 100^3 cm^3 = 1 000 000 cm^3
- And, 1 cm = 0.01 m
- Therefore, 1 cm^3 = 0.01^3 m^3 = 0.000 001 m^3

Question 12 Hard [1 mark]

Which row shows a reasonable estimate?

	quantity	value
A	kinetic energy of a bus travelling on an expressway	30 000 J
B	power of a domestic light bulb	300 W
C	temperature of a hot oven	300 K
D	volume of air in a car tyre	0.03 m^3

A

B

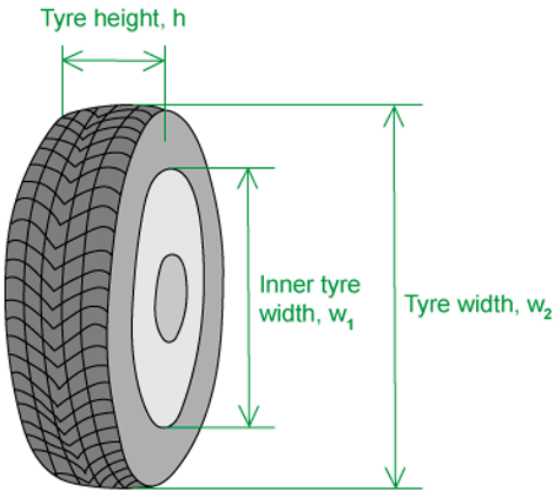
C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Modelling the tyre as a hollow cylinder with inner width w_1, outer width w_2, and height h
  - Volume of a cylinder: $V = \pi r^2 h$
  - The air occupies the outer width of the tyre only
  - Volume of air in tyre = total volume of tyre - volume of inner tyre
  - $V = \pi h \frac{1}{2} w_2{}^2 - \pi h \frac{1}{2} w_1{}^2$
  - $V = \frac{\pi h}{4} w_2^2 - w_1^2$
- Some reasonable estimates are: h ~ 15 cm, w_1 ~ 30 cm and w_2 ~ 60 cm
- Volume of air in tyre $\approx \frac{\pi \times 0.15}{4} \; 0.6^2 - 0.3^2 = 0.03 \; \mathrm{m}^3$



A is incorrect because a bus weighs at least 10^4 kg and would be travelling at least 10 m s^{-1}. Therefore, its kinetic energy would be around 10^5 - 10^6 J, which is 1 or 2 orders of magnitude greater than 30 000 J.

B is incorrect because the standard wattage of a domestic light bulb is 60 W

C is incorrect because 300 K in Celsius is 300 - 273 = 27 °C, whereas a hot oven would be at least 200 °C, which is 200 + 273 = 473 K

Question 13 Hard [1 mark]

Which statement is **incorrect** by a factor of 100 or more?

A Atmospheric pressure is about 1×10^5 Pa

B Light takes 5×10^2 s to reach us from the Sun

C The frequency of ultraviolet light is 3×10^{12} Hz

D The life-span of a man is about 2×10^9 s

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The wavelength of UV light is about 10nm
- Using the wave equation $c = f \lambda$
- $f = \frac{3 \times 10^8}{10 \times 10^{-9}} = 3 \times 10^{16}$ Hz
- *This is incorrect by a factor of 10^4*

A is incorrect because statement is correct, 1 atm is equal to 1×10^5 Pa

B is incorrect because the time it takes for light to reach the Earth from the Sun is calculated by

$$\frac{1AU}{\text{Speed of light}} = \frac{1.5 \times 10^{11}}{3 \times 10^8} = 500s = 5 \times 10^2 \text{ s}$$

D is incorrect because if we calculate the life-span of the person in the question we get $\frac{2 \times 10^9}{365 \times 24 \times 60 \times 60}$ = 63 years, which is a reasonable estimation

Question 14 Hard [1 mark]

An adult person jumps from a wall which is 2 m high. They land without rebounding.

What is the best estimate for the change in momentum of the person?

A 4×10^0 kg m s^{-1}

B 4×10^1 kg m s^{-1}

C 4×10^2 kg m s^{-1}

D 4×10^3 kg m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The person jumps from rest and lands with maximum velocity
- There is no rebound, so all the gravitational potential energy from the jump is transferred to kinetic energy
  - $E_p = mgh = m \times 10 \times 2 = 20m$
  - $E_p = E_k$
  - $20m = \frac{1}{2}mv^2$
  - $v^2 = \frac{2 \times 20m}{m}$
  - $v = \sqrt{40} \approx 6$ m s^{-1}
- The person's change in momentum is:
  - Change in momentum = final momentum – initial momentum
  - $\Delta p = mv - u$
  - $\Delta p = m6 - 0 = 6m$
- Use the answer options to find an estimate for the mass of the person:
- Option **C**: 4×10^2 kg m s^{-1} = $6m$
  - Rearrange to calculate mass: $m = \frac{4 \times 10^2}{6}$
  - So, $m = 66.7 \approx 70$ kg
- This is a suitable estimate for the mass of an adult person, so the correct answer is **C**, 4×10^2 kg m s^{-1}

A is incorrect as $m = \frac{4 \times 10^0}{6} \approx 0.7$ kg is too low an estimate for the mass of a person.

B is incorrect as $m = \frac{4 \times 10^{1}}{6} \approx 7$ kg is too low an estimate for the mass of a person.

D is incorrect as $m = \frac{4 \times 10^{3}}{6} \approx 700$ kg is too high an estimate for the mass of a person.

Question 15 Hard [1 mark]

The diameter of a proton is approximately 1×10^{-15} m.

Which estimate is reasonable for the time taken for light to cross the nucleus of a hydrogen atom?

A 3×10^{-28} s

B 3×10^{-24} s

C 3×10^{-20} s

D 3×10^{-16} s

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The speed of light in free space, $c = 3.00 \times 10^{8}$ m s^{-1} (from the data page)
- $v = \frac{s}{t} \rightarrow t = \frac{s}{v} \rightarrow t = \frac{s}{c}$
- A hydrogen nucleus contains one proton
  - A reasonable estimate for the diameter of a hydrogen nucleus is 1×10^{-15} m
- Therefore, $t = \frac{s}{c} \approx \frac{1 \times 10^{-15}}{3.00 \times 10^{8}} \approx 3.3 \times 10^{-24}$ s

Question 16 Hard [1 mark]

The bottom of the Mariana Trench is estimated to be approximately 11 000 m below sea level.

What is the best estimate of the total pressure at the bottom of the Mariana Trench?

A 1×10^6 Pa

B 1×10^7 Pa

C 1×10^8 Pa

D 1×10^9 Pa

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Hydrostatic pressure = $\rho g h$
  - A reasonable estimate for the density ρ of sea water is 1000 kg m^{-3}
  - A reasonable estimate for the gravitational field strength g on Earth is 10 N kg^{-1}
  - The depth h of the Mariana Trench is given as 11 000 m
- So, hydrostatic pressure $\approx 1000 \times 10 \times 11\ 000 \approx 1.1 \times 10^8$ Pa
- Total pressure = hydrostatic pressure + atmospheric pressure
- Atmospheric pressure is approximately 100 kPa ≈ 1×10^5 Pa
  - This is much smaller than the hydrostatic pressure, so it can be ignored in this case
- Therefore, total pressure $\approx 1 \times 10^8$ Pa
  - This is answer **C**

Question 17 Hard [1 mark]

A satellite orbits Earth at an altitude of 400 km above the Earth's surface.

What is a reasonable estimate of the total distance travelled by the satellite in one complete orbit?

A 4×10^5 m

B 4×10^6 m

C 4×10^7 m

D 4×10^8 m

Mark Scheme & Explanation:

The correct answer is C

- Circumference of orbit = $2\pi r$
- A reasonable estimate for the radius of the Earth is $6400\ \text{km} = 6.4 \times 10^6\ \text{m}$
- The altitude is given as $400\ \text{km} = 4 \times 10^5\ \text{m}$
- Therefore, orbital radius

$$r \approx 6.4 \times 10^6 + 0.4 \times 10^6 \approx 6.8 \times 10^6\ \text{m}$$

- π is ≈ 3
- Therefore

$$\text{circumference} \approx 2 \times 3 \times 6.8 \times 10^6 \approx 4 \times 10^7\ \text{m}$$

- This is answer **C**

A is incorrect as this would require an orbital radius approximately 100 times smaller than the actual value

B is incorrect as this underestimates the orbital radius or uses an incorrect formula

D is incorrect as this overestimates the orbital radius by a factor of 10